

for versioning and namespacing of transaction logic.

- **repeated string inputs and repeated string outputs:** Lists of state addresses that this transaction intends to read from (inputs) and write to (outputs). This is crucial for concurrency control and ensuring transactions do not interfere with each other's state changes.
- **string nonce:** A unique value used once to ensure that each transaction's header is unique, preventing replay attacks.
- **string payload_encoding:** Specifies how the payload is encoded, ensuring that the transaction processor knows how to interpret it.
- **string payload_sha512:** A SHA-512 hash of the payload, ensuring integrity and allowing for validation without revealing the payload's content before necessary.
- **string signer_public_key:** The public key of the individual or entity that signed the transaction, ensuring authenticity and non-repudiation.

The connection between transactions and batches is shown in Figure 3.

Writing transactions and batches

To write transactions and batches in Sawtooth, follow these general steps.

- **Define transaction payload:** Structure the payload according to the rules of the transaction family your application is interacting with.
- **Create transaction:** Construct the transaction header with appropriate metadata, including inputs, outputs, and the family name and version. Sign the header with your private key to generate the header signature, and assemble the transaction with the payload and signed header.
- **Group transactions into batches:** If you have multiple transactions, organise them into batches. Create a batch header with the list of transaction signatures and sign it. Combine the transactions and the

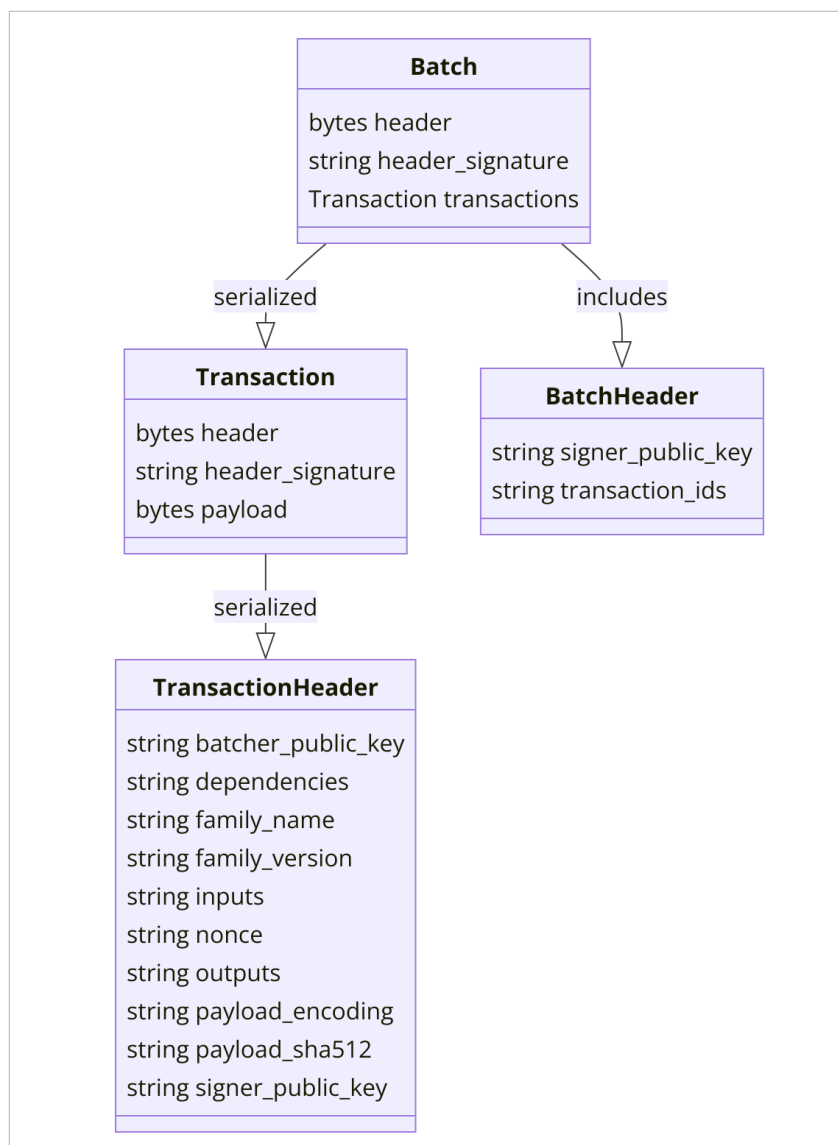


Figure 3: Transactions and batches

- signed header to form a batch.
- **Submit to the network:** Use the Sawtooth REST API or an SDK in your preferred programming language to submit the batches to a Sawtooth validator. This process involves specific coding practices and interactions

with the Sawtooth SDKs, which vary depending on the programming language you're using (Python, JavaScript, etc). Sawtooth documentation provides detailed guides and examples for creating and submitting transactions and batches using these SDKs. END 🐧

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